

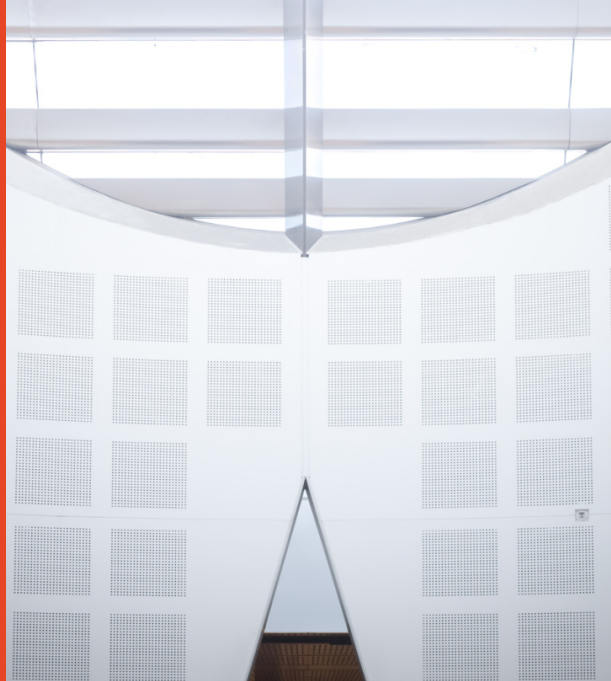
Solving Optimal Stopping Problems with Reinforcement Learning

MMathSci Thesis Presentation

Christopher Morton

Supervised by Dr. Zhou Zhou

23rd April, 2026



What is Optimal Stopping?

In optimal stopping, we find the best time τ to **perform an action**, maximising:

$$\sup_{\tau} \mathbb{E}[e^{-\alpha\tau} X_{\tau}]$$

- ▶ X : **Stochastic process** - specifies reward over time.
- ▶ α : **Discounting rate** - incentivises stopping.
- ▶ τ : **Stopping time** - represents time to act.

X is a risky process, but at time τ we lock in the reward.

What is Optimal Stopping?

In optimal stopping, we find the best time τ to **perform an action**, maximising:

$$\sup_{\tau} \mathbb{E}[e^{-\alpha\tau} X_{\tau}]$$

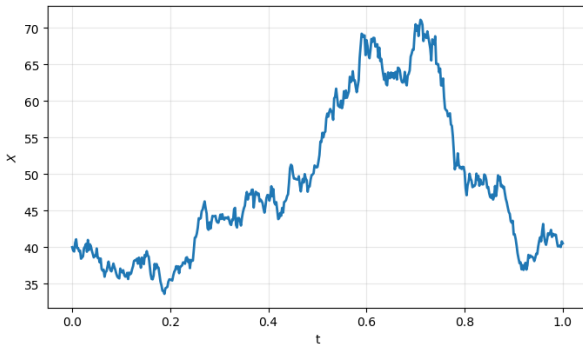
There are several real-world problems involving optimal stopping:

- ▶ **Mathematical finance:** Asset pricing, portfolio optimisation.
 - ▶ X : stock price, τ : time to sell.
- ▶ **Engineering:** Sequential detection, signal processing.
 - ▶ X : observed signal, τ : time to raise alarm.

What is Optimal Stopping?

Suppose X is a stock price following $dX_t = b(X_t)dt + \sigma(X_t)dW_t$.

- ▶ You buy at $X_0 = 40$. What is the optimal time to sell?



What if we don't know the underlying dynamics of X ?

What is Reinforcement Learning?

Reinforcement learning develops effective strategies by **learning from itself**.

- ▶ Makes **no assumptions** on model parameters.
- ▶ Updates strategy after each observation.

Reinforcement learning strikes a balance between:

- ▶ **Exploitation**: Maximise rewards based on current model.
- ▶ **Exploration**: Try new actions to improve model accuracy.

Applying Reinforcement Learning to Optimal Stopping

Optimal stopping

- ▶ Precise mathematical formulation.
- ▶ Clear notion of optimality.
- ▶ Requires specified model.

Reinforcement learning

- ▶ Learns policies from data.
- ▶ Model-free algorithm.
- ▶ Scales to complex settings.

In this thesis we develop a reinforcement learning framework for several classes of optimal stopping problems.

Applying Reinforcement Learning to Optimal Stopping

Classical optimal stopping: Pure stopping time τ .

$$\sup_{\tau} \mathbb{E}[e^{-\alpha\tau} g(X_{\tau})]$$

- ▶ Example: Stop when $X > 50$.

Randomised optimal stopping: Stopping intensity π_t .

$$\sup_{\pi} \mathbb{E} \left[\int_0^T e^{-\alpha t} g(X_t) \pi_t p_t dt + e^{-\alpha T} g(X_T) p_T \right]$$

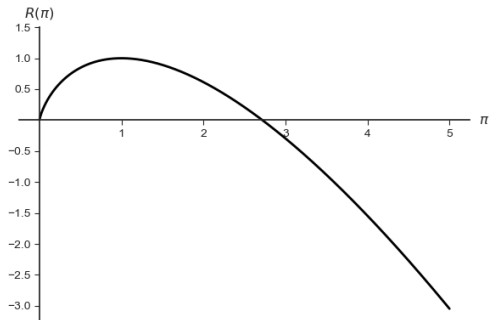
- ▶ p_t : Probability of survival to time t .
- ▶ $\pi_t p_t dt$: Probability of stopping over $[t, t + dt]$.
- ▶ Example: Stop with higher probability when $X > 50$.

Applying Reinforcement Learning to Optimal Stopping

Problem: These problems are identical!

► Optimal intensity: $\pi_t = \infty$ when $t \geq \tau^*$, $\pi_t = 0$ otherwise.

Solution: Introduce entropy $R(\pi) = \pi - \pi \log \pi$ to encourage exploration.



► $R(0) = 0$, $R(\pi) \rightarrow -\infty$ as $\pi \rightarrow \infty$.

► Maximised at $\pi = 1$.

Applying Reinforcement Learning to Optimal Stopping

This gives us the **entropy-regularised objective**:

$$\sup_{\pi} \mathbb{E} \left[\int_0^T e^{-\alpha t} (g(X_t) \pi_t p_t + \lambda R(\pi_t) p_t) dt + e^{-\alpha T} g(X_T) p_T \right]$$

- ▶ λ : Exploration parameter - larger λ promotes exploration.

How do we solve the entropy-regularised problem?

Results

Recall the entropy-regularised objective:

$$V^\pi(t, x) = \mathbb{E} \left[\int_t^T e^{-\alpha(s-t)} (g(X_s) \pi_s p_s + \lambda R(\pi_s) p_s) ds + e^{-\alpha(T-t)} g(X_T) p_T \right]$$

Theorem (HJB Equation and Optimal Policy)

For a fixed intensity π , the value $V^\pi(t, x)$ solves the PDE

$$(\partial_t + \mathcal{L})V^\pi(t, x) - \alpha V^\pi(t, x) + (g(x) - V^\pi(t, x))\pi(t, x) + \lambda R(\pi(t, x)) = 0,$$

with terminal condition $V^\pi(T, x) = g(x)$.

Taking the supremum over $\pi \geq 0$ (with $V^ = \sup_{\pi} V^\pi$) yields the maximiser*

$$\pi^*(t, x) = \exp \left(-\frac{V^*(t, x) - g(x)}{\lambda} \right).$$

Results

Recall the entropy-regularised objective:

$$V^\pi(t, x) = \mathbb{E} \left[\int_t^T e^{-\alpha(s-t)} (g(X_s) \pi_s p_s + \lambda R(\pi_s) p_s) ds + e^{-\alpha(T-t)} g(X_T) p_T \right]$$

Theorem (Monotone Policy Iteration)

Define (π_n, V^{π_n}) iteratively by:

- ▶ **Policy evaluation:** given π_n , solve the linear PDE for V^{π_n} .
- ▶ **Policy update:** set $\pi_{n+1}(t, x) = \exp\left(-\frac{V^{\pi_n}(t, x) - g(x)}{\lambda}\right)$.

Then:

$$V^{\pi_n} \uparrow V^* \quad \text{and} \quad \pi_n \downarrow \pi^* \quad \text{as } n \rightarrow \infty.$$

Using Reinforcement Learning in Practice

In practice, we do not solve the PDE exactly. We instead approximate V with a parametric model V_θ and train it using regression.

$$\pi_1 \longrightarrow V_\theta^{\pi_1} \longrightarrow \pi_2 \longrightarrow V_\theta^{\pi_2} \longrightarrow \dots \longrightarrow \pi^* \longrightarrow V_\theta^*$$

For our numerical examples, we use a fully connected neural network to learn the model from simulated data alone and compute the relative error to the theoretical value.

Numerical Example - American Put

We use the policy iteration algorithm on the classical American put

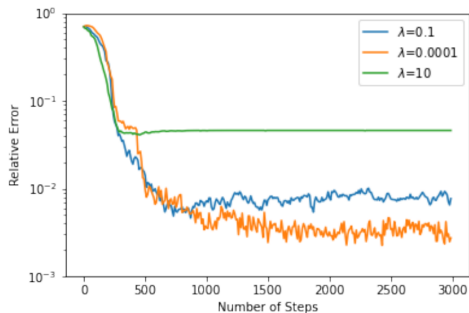
$$g(X) = (K - X)^+, \quad dX_t = X_t(r dt + \sigma dW_t)$$

with parameters

$$X_0 = K = 40, \quad r = 0.06, \quad \sigma = 0.40, \quad T = 1.$$

Interpretation: The holder has the right to sell the asset X to the issuer at any time for the agreed strike price K .

Numerical Example - American Put



Adjusting λ gives us a tradeoff:

- ▶ Larger λ accelerates convergence.
- ▶ Smaller λ reduces bias.

Summary & Further Extensions

The randomised method provides a consistent way to solve optimal stopping problems when the underlying dynamics are unknown and must be learned from data.

The intensity-based viewpoint provides a scaffold for several extensions:

- ▶ **Double stopping:** Problems that require multiple exercise times.
- ▶ **Time-inconsistent stopping:** Discounting is no longer exponential.
- ▶ **Adaptive exploration:** Allow λ to vary over training.

Double Stopping Problem

In the double stopping problem choose two stopping times (τ_1, τ_2) to maximise

$$v(S) = \operatorname{ess\,sup}_{\tau_1, \tau_2 \in \mathcal{T}_{[S, T]}} \mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_S].$$

Theorem (Reduction Principle)

For each θ , define

$$u_1(\theta) = \operatorname{ess\,sup}_{\tau_1 \geq \theta} \mathbb{E}[\psi(\tau_1, \theta) \mid \mathcal{F}_\theta], \quad u_2(\theta) = \operatorname{ess\,sup}_{\tau_2 \geq \theta} \mathbb{E}[\psi(\theta, \tau_2) \mid \mathcal{F}_\theta],$$

and set $\phi(\theta) = \max\{u_1(\theta), u_2(\theta)\}$. Then the single-stopping value

$$u(S) = \operatorname{ess\,sup}_{\theta \geq S} \mathbb{E}[\phi(\theta) \mid \mathcal{F}_S]$$

satisfies $v(S) = u(S)$ almost surely.

References

- [1] Zafarali Ahmed et al. “Understanding the impact of entropy on policy optimization”. In: *International conference on machine learning*. PMLR. 2019, pp. 151–160.
- [2] Yuchao Dong. “Randomized Optimal Stopping Problem in Continuous Time and Reinforcement Learning Algorithm”. In: *SIAM Journal on Control and Optimization* 62.3 (2024), pp. 1590–1614. DOI: 10.1137/22M1516725.
- [3] Magdalena Kobylanski, Marie-Claire Quenez, and Elisabeth Rouy-Mironescu. “Optimal double stopping time problem”. en. In: *Comptes Rendus. Mathématique* 348.1-2 (2010), pp. 65–69. DOI: 10.1016/j.crma.2009.11.020.
- [4] Wenpin Tang, Yuming Paul Zhang, and Xun Yu Zhou. “Exploratory HJB equations and their convergence”. In: *SIAM Journal on Control and Optimization* 60.6 (2022), pp. 3191–3216.